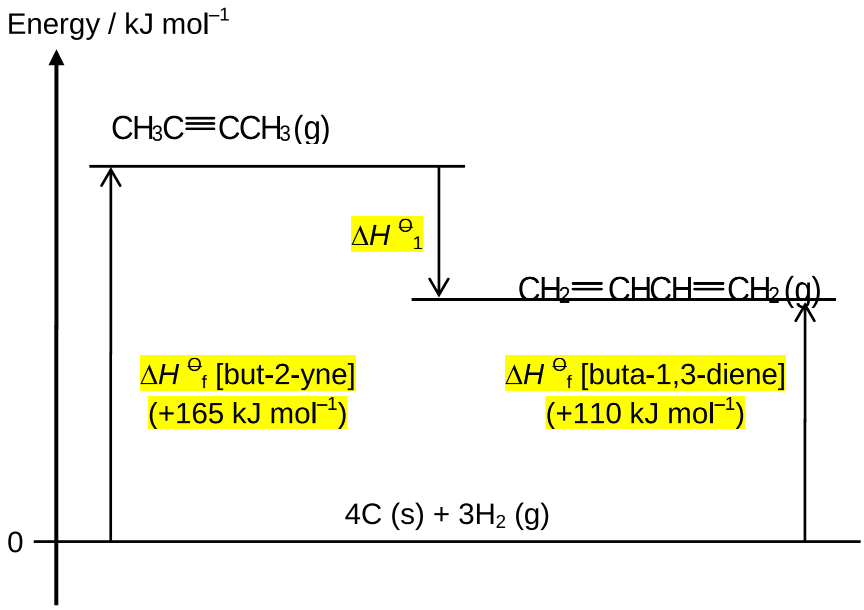
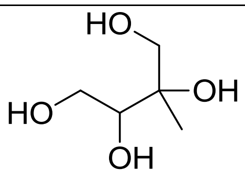
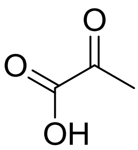
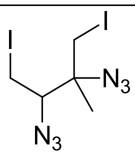
H1 2013 Preliminary Examination Paper 2 (Suggested Solutions)

Section A

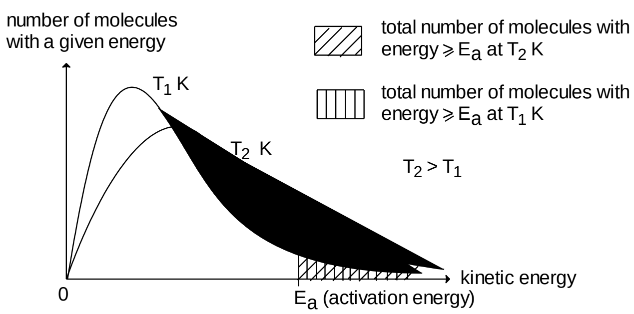
A1(a)(i)	Mg: $1s^2 2s^2 2p^6 3s^2$ A*l*: $1s^2 2s^2 2p^6 3s^2 3p^1$	
(ii)	The valence electron to be removed from an A*l* atom is in the 3p subshell while that from a Mg atom is in the 3s subshell. The 3p subshell is further from the nucleus [1/2] than the 3s subshell. Valence electron in A*l* experiences less electrostatic attraction [1/2] than the valence 3s electron in Mg. Thus, less energy is required to remove an electron in the 3p subshell than to remove an electron in the 3s subshell.	**[2]**
	Marker's comments: The correct terms should be used! E.g. "subshell"	
(b)	MgO is an ionic oxide which is basic and reacts with acids to form salts and water. $MgO + 2H^+ \rightarrow Mg^{2+} + H_2O$ Al_2O_3 is an ionic oxide with covalent character in its ionic bonding. It has both acidic and basic properties. $Al_2O_3 + 2OH^- + 3H_2O \rightarrow 2Al(OH)_4^-$ Al^{3+} has high charge density, hence high polarising power. It distorts the electron cloud of O^{2-} such that there is covalent character to the Al-O interaction.	**[4]**
	Marker's comments: Candidates must relate the acid-base properties of the oxides to the bonding. Aluminium oxide reacts with both acids and bases as its bonding is predominantly ionic with covalent character – a brief explanation of the covalent character in its bonding is required.	
(c)(i)	$x = 8$ [1]	
(ii)	$Mg_2Si_3O_8 + 4HCl \rightarrow 2MgCl_2 + 3SiO_2 + 2H_2O$ [1]	**[2]**
(d)(i)	Calcium chloride and magnesium sulfate are ionic compounds which exist as ions in the molten state. [1/2] These mobile ions can act as mobile charged carriers [1/2] when an electric current is applied, and delivered into the skin via the pores.	
	Marker's comments: To explain why they are conductors, there must be mention of mobile charge carriers. Many students mistakenly talked about "delocalised electrons" – which are the mobile conductors in metallic structures, not in ionic structures like calcium chloride and magnesium sulfate.	
(ii)	Quote data from Data Booklet The ionic radius of Ca^{2+} is greater than that of Mg^{2+}. Ca^{2+} ions has 1 additional filled principal quantum shell than Mg^{2+} ions. The valence electrons in Ca^{2+} ions are increasingly further away from the nucleus and are less strongly attracted to the nucleus than in Mg^{2+}.	**[3]**
	Marker's comments: The correct terminology must be used! The ionic sizes from the Data Booklet must be quoted in your answer.	

A2(a)	The standard enthalpy change of formation is the **enthalpy change** when **1 mole of a pure compound** in a specified state is formed from **its constituent elements in their standard states**, at **298 K and 1 atm**.	**[1]**
(b)(i)		
(ii)	Labelling of $\Delta H^{\ominus}_1$ on diagram with correct direction of arrow [By Hess' Law, $165 + \Delta H^{\ominus}_1 = 110$ $\Delta H^{\ominus}_1 = -55$ kJ mol^{-1}	
(iii)	Buta-1,3-diene is **more stable** than but-2-yne since the **p-orbitals of all the C atoms** in buta-1,3-diene **are able to overlap**, thus allowing the delocalisation of π electrons to occur throughout all 4 C atoms. [2]	
	Marker's comments Candidates do not seem to understand how "delocalisation" occurs. Delocalisation of π electrons occurs in $CH_2{=}CHCH{=}CH_2$ as the 4 p-orbitals used for π-bonding (for 2 π bonds) are adjacent to each other. In other words, this delocalisation does not occur in penta-1,4-diene $CH_2{=}CHCH_2CH{=}CH_2$, as the 4 p-orbitals are not adjacent to each other but separated by a $-CH_2$ group.	
(iv)	The enthalpy change of combustion of buta-1,3-diene is less exothermic than that of but-2-yne.	**[6]**
(c)(i)	$CH_3C{\equiv}CCH_3$ (g) + $2H_2$ (g) $\longrightarrow$ $CH_3CH_2CH_2CH_3$ (g)	
	Marker's comments: Question states that "excess" hydrogen is used. Hence the C≡C bond is reduced to C-C and not C=C.	

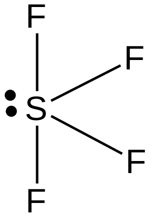
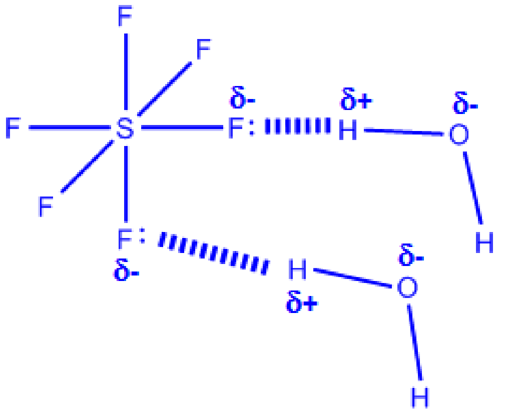
(ii)	Use of Ni/Pt/Pd catalyst	
(iii)	$\Delta H^{\ominus}_{hydrogenation}$ [but-2-yne] = [E(C≡C) + 2E(H–H)] – [E(C–C) + 4E(C–H)] = [(840) + 2(436)] – [(350) + 4(410)] = –278 kJ mol^{-1} [2]	
	Marker's comments: Many candidates missed out on the bond energy of H-H bond.	
(iv)	$\Delta H^{\ominus}_{f}$ [butane] = +165 – 278 = –113 kJ mol^{-1} [1]	**[5]**

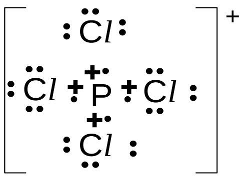
A3(a)(i)	 A black precipitate of MnO_2 is observed.	
	Marker's comments: Many candidates do not seem to know the difference between using HOT $KMnO_4$ and COLD $KMnO_4$. Cold $KMnO_4$ causes mild oxidation and a diol is formed, where 2 –OH groups add to a C=C bond. Many did not read the question properly as well. The observation of "$KMnO_4$ is decolorised" has already been stated in the question, hence it should not be repeated in the answer.	
(ii)	 Effervescence. colourless gas (CO_2), which forms a white ppt with $Ca(OH)_2$(aq) is evolved.	**[4]**
	Marker's comments: Candidates did not know what are "organic" products – organic compounds contain both C & H atoms. Hence CO_2 and H_2O are not organic products and should not be drawn in the answer. An observation should be a visible change – hence instead of just saying that CO_2 is formed in the observation, "effervescence" and a positive test for the gas should be stated.	
(b)(i)		
(iii)	C*l*–N bond would have a higher bond energy than I–N since the C*l* orbital involved in bonding is smaller and thus less diffuse compared to that of I, resulting in a more effective overlap.	**[3]**

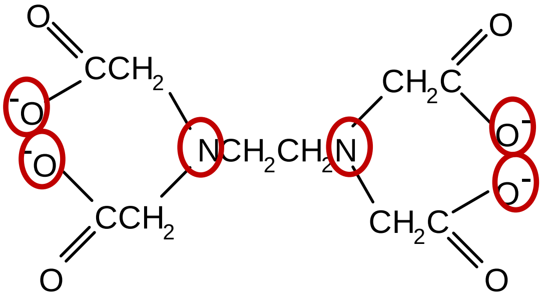
	Marker's comments: In comparing bond strengths, it is important to use the correct terms: Cl orbital is less "diffuse"; "orbital overlap" is more effective".	

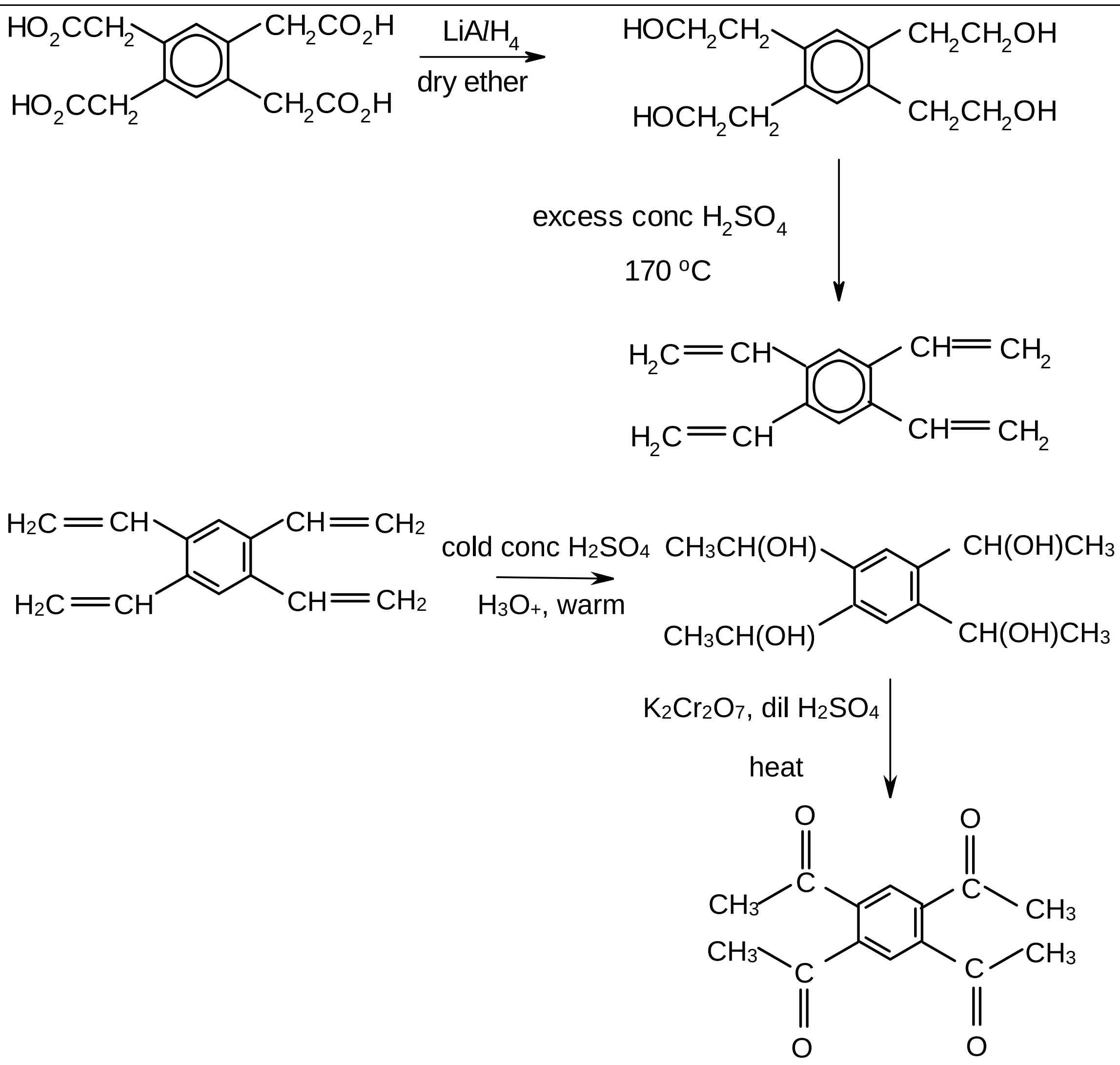
A4(a)	Simple molecular structure with weak intermolecular van der Waals' forces	**[1]**
	Marker's comments: It is stated in the question that As_2O_3 reacts with **alkalis**. This means that As_2O_3 is acidic, and hence its bonding is covalent. Since As is not a group IV element like Si, the structure is not giant covalent structure.	
(b)	Mass of As needed = 2 × 60 = 120 mg Amount of As needed = $(120 \times 10^{-3})/ 74.9 = 1.602 \times 10^{-3}$ mol Amount of As_2O_3 needed = $(1.602 \times 10^{-3})/ 2 = 8.010 \times 10^{-4}$ mol [1] Mass of As_2O_3 needed = $8.010 \times 10^{-4} \times 197.8 = 0.158$ g or 158 mg [1]	**[2]**
(c)(i)	12 ppb [1]	
(ii)	Concentration of As in urine = $\frac{12}{\frac{1}{2}\frac{11}{4}}$ = 80.7 ppb [1]	**[2]**
(d)(i)	Redox reaction [1]. AsO_3^{3-} is oxidised as the oxidation number of As increases from +3 to +5 [1/2] and I_2 is reduced as the oxidation number of I decreases from 0 to -1 [1/2].	
	Marker's comments: The question states to deduce the type of reaction "based on changes in oxidation number". Hence the changes in oxidation no. should be stated in answer.	
(ii)	***The distribution of molecular energies at two different temperatures.***  Diagram An increase in temperature from T_1 to T_2 K increases the average kinetic energy of the reactant molecules . As such at the higher temperature, significantly more reactant molecules have energy greater than or equal to the activation energy of the reaction (as shown by the larger shaded area in the above diagram). This results in an increase in the effective collision frequency and hence an increase in the rate of the reaction.	**[5]**

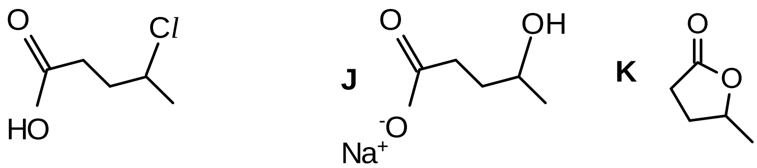
Year 6 H1 2011 Prelim Paper 2 - Section B

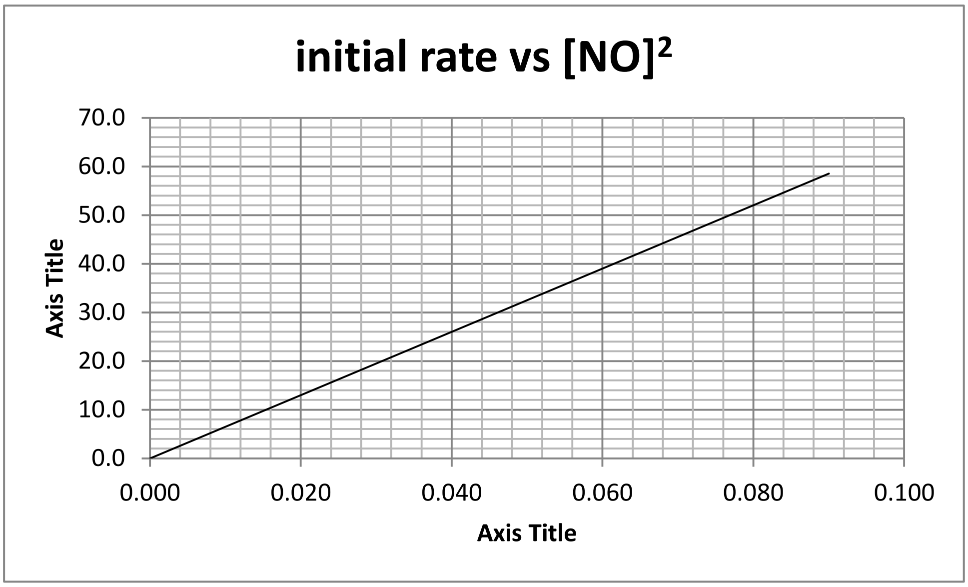
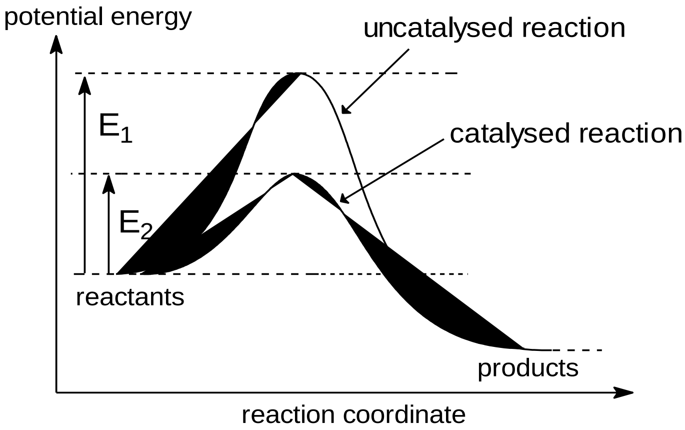
B5(a)(i)		[1]	
(ii)	There are 4 bond pairs and 1 lone pair around the S atom. To minimise repulsion, the 5 electron pairs are directed to the corners of a trigonal bipyramid. Bond pair–lone pair repulsion > bond pair–bond pair repulsion Hence shape is distorted tetrahedral/ see-saw/ sawhorse.	[2]	
(iii)		[2]	**[5]**
(b)(i)	Mass of HNO_3 = $\frac{(156 \times 10^6) \times 1.5}{1 \times 10^6}$ = 234 g No. of moles of HNO_3 (in 156×10^6 cm^3 of water) = $\frac{234}{63}$ = 3.714 mol $[HNO_3]$ = $\frac{3.714}{156 \times 10^6}$ x 1000 = 2.38×10^{-5} mol dm^{-3}	[2]	
(ii)	$CaCO_3 + 2HNO_3 \rightarrow Ca(NO_3)_2 + CO_2 + H_2O$	[1]	
(iii)	No. of moles of HNO_3 that came into contact with statue = $\frac{0.1}{100} \times 3.714 = 3.714 \times 10^{-3}$ $CaCO_3 \equiv 2HNO_3$ Mass of statue lost = $(3.714 \times 10^{-3}) \times 100.1 \times 0.5 = 0.186$ g	[1]	**[4]**

(c)(i)	A catalyst is required to speed up the reaction, since the activation energy required for the reaction is high. [1]	[1]	
(ii)	When pressure is increased, by Le Chatelier's Principle, the system at equilibrium favours the side with less gaseous particles to decrease pressure, i.e. position of equilibrium shifts to the right. Hence yield of SO_3 increases. [1]	[1]	
(iii)	$\Delta H^{\ominus} = -396 - (-297) = -99$ kJ mol^{-1}	[2]	**[4]**
(d)	PCl_4^+: Since it has 4 electron-pairs with no lone pairs, its shape is tetrahedral. 	[1] [1]	**[2]**
(e)(i)	When a few drops of water is added, <u>partial hydrolysis</u> occurs forming a <u>white solid</u> <u>$POCl_3$</u> is formed with <u>white HCl fumes</u>. PCl_5 (s)+ H_2O (l) $\longrightarrow$ $POCl_3$ (s) + $2HCl$ (g) When large amount of water is added, PCl_5 undergoes <u>complete hydrolysis</u> to form a very acidic solution consisting of H_3PO_4 and HCl. PCl_5 (s)+ $4H_2O$ (l) $\longrightarrow$ H_3PO_4(aq) + $5HCl$(aq) When a few drops of UI is added, the colour is red.	[0.5] [1] [1] [0.5]	
(ii)	From $SOCl_2$: H_2SO_3 ; oxidation number of S = +4 From SO_2Cl_2 : H_2SO_4 ; oxidation number of S = +6	[1] [1]	**[5]**
			[Total: 20]

B6(a)(i)	Amide [1]	[1]	
(ii)	$EDTA^{4-}$ contains lone pairs of electrons which can be donated into the low-lying empty orbitals of Mn^{2+} [1]	[1]	
(iii)	 [1]	[1]	
(iv)	$[Mn(EDTA)]^{2-}$ [1]	[1]	

(b)	 HO2CCH2 CH2CO2H HO2CCH2 CH2CO2H LiAlH4 dry ether HOCH2CH2 CH2CH2OH HOCH2CH2 CH2CH2OH excess conc H2SO4 170 °C H2C=CH CH=CH2 H2C=CH CH=CH2 H2C=CH CH=CH2 H2C=CH CH=CH2 cold conc H2SO4 H3O+, warm CH3CH(OH) CH(OH)CH3 CH3CH(OH) CH(OH)CH3 K2Cr2O7, dil H2SO4 heat O C CH3 O C CH3 CH3 C O CH3 C O [6]	**[10]**

(c)	① **G** produces effervescence with Na_2CO_3. ⇒ **G** is a carboxylic acid. [½] ② **G** undergoes nucleophilic substitution and oxidation with alkaline I_2. ⇒ **G** contains the structure $\begin{array}{l} CH_3CH— \\ \;\;\;\;\;\;\;\;\; \vert \\ \;\;\;\;\;\;\; Cl \end{array}$ [½] ③ **G** undergoes nucleophilic substitution with NaOH(aq) to form **J**. ⇒ **J** contains the structure $\begin{array}{l} CH_3CH— \\ \;\;\;\;\;\;\;\;\; \vert \\ \;\;\;\;\;\;\; OH \end{array}$ [½] ④ **G** is resistant to oxidation. ⇒ A does not contain alkenes, primary or secondary alcohol, or aldehyde. [½] After acidification, **J** undergoes condensation reaction with hot conc H_2SO_4 to form **K**. **K** does not undergo electrophilic addition with aqueous Br_2. ⇒ **K** is an ester. [½] [max 2m] **G** **J** **K**  [1m x 3]	[2] [3]	[5]
(d)(i)	A system is said to have reached dynamic equilibrium when the rates of forward and backward reactions are equal and non-zero [1] and there is no change in the concentrations of the reactants and products.	[1]	
(ii)	$K_c = \frac{[CH_3CH_2CH(OH)_2]}{[CH_3CH_2CHO][H_2O]}$	[1]	
(iii)	$K_c = \frac{0.324}{0.676 \, x \, 0.676} = 0.709 \text{ mol}^{-1} \text{ dm}^3$ computation [½], units: [½]	[1]	
(iv)	The value of K_c for the reaction between propanone and water is smaller than that for the reaction between propanal and water. $CH_3COCH_3(l) + H_2O(l) \rightleftharpoons (CH_3)_2C(OH)_2(l)$ ……….(1) $CH_3CH_2CHO(l) + H_2O(l) \rightleftharpoons CH_3CH_2CH(OH)_2(l)$ ………(2) The position of equilibrium of (2) lies more to the right hand side, as compared to (1) [1]. Equilibrium (2) favours the formation of the product to a larger extent. Thus, CH_3CH_2CHO is more reactive than CH_3COCH_3 [1].	[2]	[5]
			[Total: 20]

B7(a)(i)	 Appropriate scale etc [0.5] Best fit line through origin [0.5] When the graph of initial rate against $[NO]^2$ is plotted, a straight line graph passing through the origin is obtained. Hence initial rate is proportional to $[NO]^2$ $\Rightarrow$ reaction is second order with respect to [NO]. [1]	[3]	
(ii)	rate = $k[H_2][NO]^2$ [1]	[1]	
(iii)	Subst. initial rate = 6.5×10^{-4} mol dm^{-3} s^{-1}, [NO] = 0.0100 mol dm^{-3}, $[H_2]$ = 0.50 mol dm^{-3}, $6.5 \times 10^{-4} = k(0.50)(0.0100)^2$ k = 13 mol^{-2} dm^6 s^{-1} [1] correct answer + units	[1]	
(iv)	**Energy Profiles for a Catalysed and Uncatalysed Reaction**  E_1 = activation energy of uncatalysed reaction E_2 = activation energy of catalysed reaction [2]	[2]	**[7]**

(b)(i)	Step I – Cl_2 in the dark Step II – KCN in ethanol, heat under reflux **A**: CH_2CH_2 | | CN CN	[1] [1] [1]	
(ii)	Type of reaction in step II – nucleophilic substitution Equation: $CH_2(Cl)CH_2Cl + 2CN^- \rightarrow CH_2(CN)CH_2CN$	[1] [1]	
(iii)	Warm each compound with a solution containing $AgNO_3$ in excess $NH_3(aq)$ (Tollens' reagent). [1] Only P gives silver mirror. [1] OR Warm each compound with complex copper(II) ions in alkaline solution (Fehling's solution). Only P gives reddish-brown precipitate. OR Heat each compound with dilute H_2SO_4 and $K_2Cr_2O_7(aq)$. Only P turns the orange solution green. OR Add 2,4-DNPH to each compound. Only P gives an orange solid.	[2]	**[7]**
(c)(i)	pH of HCl solution = $-\lg 0.0100 = 2$ [1]	[1]	
(ii)	$K_a = 10^{-4.88} = 1.318 \times 10^{-5}$ mol dm^{-3} $[H^+]=\sqrt{K_a \times c}=\sqrt{1.318\times10^{-5}\times0.0100}= 3.630\times10^{-4}$ mol dm^{-3} [1] $pH = -\log(3.36 \times 10^{-4}) = 3.44$ [1]	[1] [1]	
(iii)	HCl solution has lower pH than 2-methylpropanoic acid of the same concentration, as HCl is a <u>strong acid which is fully dissociated to give H^+.</u> Hence $[H^+] = [HCl]$ Whereas 2-methylpropanoic acid has a higher pH as it is a <u>weak acid that is only partially dissociated</u>. Hence $[H^+] << [acid]$	[0.5] [0.5]	
(iv)	3-hydroxybutanoic acid has a lower pK_a value than 2-methylpropanoic acid, indicating that it is a stronger acid. [1] The anion $CH_3CH(OH)CH_2COO–$ formed from the dissociation of 3-hydroxybutanoic acid is more stable [1/2] than the anion $CH_3CH(CH_3)COO^-$ from dissociation of 2-methylpropanoic acid. This is due to the electron-withdrawing –OH group decreasing the electron density on the carboxylate ion [1/2] , hence stabilising the anion to a greater extent. This increases the extent of dissociation of $CH_3CH(OH)CH_2COOH$ relative to $CH_3CH(CH_3)COOH$.	[2]	**[6]**
	[Total: 20]		